

Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
S. Y. B.Tech. Computer Science and Engineering
Open Elective-I and Open Elective-II
W.E.F. 2026-27



Computer Science and Engineering

B. Tech. Semester III

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		ESE	OE/ POE	ISE	ICA	
25##U3ON*4T	Open Elective I	2	--	--	2	60	--	40	--	100
25##U3ON*4A	Open Elective I (Tutorial)	--	1	--	1	--	--	--	25	25
	Total	2	1	--	3	60	--	40	25	125

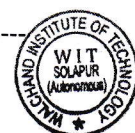
B. Tech. Semester IV

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		ESE	OE/ POE	ISE	ICA	
25##U4ON*5T	Open Elective II	2	--	--	2	60	--	40	--	100
25##U4ON*5A	Open Elective II (Tutorial)	--	1	--	1	--	--	--	25	25
	Total	2	1	--	3	60	--	40	25	125

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HEAD

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Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of offering Programme
- Internal Continuous Assessment (ICA): ICA shall be a continuous process based on the performance of the student in assignments, class tests, quizzes, attendance and interaction during theory and lab sessions, journal writing, report presentation, etc., as applicable.

List of Open Electives:**Open Electives- I (Semester-III)**

Sr. No.	Subject Code	Subject
1	25CEU3O14T	Managerial Economics
2	25MAU3O24T	Renewable Energy
3	25ETU3O34T	Sustainable Development
4	25CSU3O44T	Management Information Systems
5	25ECU3O54T	Fundamentals Of Digital Marketing
6	25ITU3O64T	Cyber Laws

Open Electives- II (Semester-IV)

Sr. No.	Subject Code	Subject
1	25GEU4O15T	Higher Engineering Mathematics
2	25GEU4O25T	Advanced Engineering Mathematics
3	25GEU4O35T	Applied Mathematics
4	25GEU4O45T	Statistics and Fuzzy logic
5	25GEU4O55T	Applied Statistics
6	25GEU4O65T	Computational and Optimization Techniques

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WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
S.Y. B. Tech. (Computer Science and Engineering)
Semester-III

25CSU3O44T: MANAGEMENT INFORMATION SYSTEMS (Open Elective-I)

Teaching Scheme

Lectures: 2 Hours/Week

Tutorial : 1 Hour/Week

Examination Scheme

ESE : 60 Marks

ISE : 40 Marks

ICA : 25 Marks

COURSE OBJECTIVES:

1. Understand the role and impact of information systems in global business and organizational strategy.
2. Examine ethical, social issues of information systems and understand IT infrastructure trends.
3. Explore information systems security and understand e-commerce dynamics.

COURSE OUTCOMES:

At the end of this course, students will be able to

1. Identify how information systems transform business processes and competitive strategies.
 2. Analyze ethical implications and evaluate IT infrastructure trends for business efficiency.
 3. Develop strategies for information system security and understand e-commerce principles.
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SECTION-I

Unit 1: Information Systems in Global Business Today

(06)

The Role of Information Systems in Business Today, How information systems are transforming business, What is new in information system. Business Processes and Information systems, Tools and technologies for collaboration and social business.

Unit 2: Information Systems, Organizations, and Strategy

(04)

Organizations and its features, How Information Systems Impact on Organizations, Competitive strategies using information systems, Challenges posed by strategic information systems

Unit 3 : Ethical and Social Issues in Information Systems

(05)

Understanding Ethical, Social, political issues raised by information systems, principles for conduct in ethical decisions, Contemporary information systems technology. Challenges to the protection individual privacy and intellectual property.

SECTION-II

Unit 4: IT Infrastructure and Emerging Technologies

(06)

IT Infrastructure, Infrastructure Components, Contemporary Hardware Platform Trends, Contemporary Software Platform Trends, Organizing Data in a Traditional File Environment, Major Capabilities of Database Management Systems, Using Databases to Improve Business Performance and Decision Making, Managing Data Resources



Unit 5: Securing Information Systems**(04)**

System Vulnerability and Abuse, Business Value of Security and Control, Organizational Framework for Security and Control, Technologies and Tools for safeguarding Information Resources

Unit 6: E-commerce: Digital Markets, Digital Goods**(05)**

Features of e-commerce, Digital Markets, Digital Goods, principles ecommerce business and revenue models, e-commerce business-to-business transaction, Role of M-commerce in business & its applications

Internal Continuous Assessment (ICA) :

Teacher should prepare a group of 4-5 students and assign them any case study based on the above chapters. These are few topics for case studies are :

1. Digital Transformation at Indian Railways
2. Data-Driven Decision Support Systems in Reliance Indu
3. Digital Health Mission and e-Hospital Systems
4. Unified Payments Interface (UPI) System
5. Reliance JioMart's E-commerce Strategy
6. SAP ERP Implementation at Tata Steel/ Infosys
7. Flipkart's Supply Chain Innovation
8. TCS's Use of TCS iON for HR Management
9. HDFC Bank's Use of CRM for Customer Engagement
10. Delhivery's Technology-Driven Logistics

Text Books:

1. Management Information Systems : Managing the Digital Firm, 15th Edition by Kenneth C.Laudon and Jane Laudon, Pearson Education
2. Management Information Systems: Sashikala Parimi, Kogent Learning Solutions Inc.

Reference Books:

1. Information Technology for Management: Transforming Organizations in the Digital Economy, Efraim Turban, 6th Edition, Wiley Edition
2. Management Information Systems: Shubhalakshmi Joshi, Smita Vaze, Biztantra





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
Second Year B. Tech.
Semester-IV

25GEU4045T – STATISTICS AND FUZZY LOGIC (Open Elective-II)

Teaching Scheme:

Lecture: 2 hr/week, 2 credits

Tutorial: 1 hr/week, 1 credit

Examination Scheme:

ESE: 60 Marks

ISE: 40 Marks

ICA: 25 Marks

Introduction:

This course introduces basic and advanced topics of statistics which are useful in data analysis. This course introduces numerical methods (Unlike analytical methods) to solve simultaneous systems of linear equations. This course also introduces the fuzzy set theory in brief which deals with characterizing the concept of uncertainty and its relationship to the increasingly important concept of information and complexity. It also includes optimization techniques like simplex method and assignment problems.

Course Prerequisite:

Elementary set theory, Basics of statistics and Elementary transformations in matrices.

Course Objectives:

1. To introduce to students the basic and advanced concepts of statistics.
2. To introduce to students various numerical methods to solve the system of linear simultaneous equations.
3. To introduce to students comprehensive coverage of fuzzy sets.
4. To enable students to solve Linear Programming Problems and Assignment Problems.

Course Outcomes:

After completing this course, student shall be able to –

1. Apply the basic concepts of statistics to interpret the variability of data.
2. Develop the relation between two variables for the given data using regression and fitting the curves for the given data.
3. Solve the system of linear simultaneous equations arising in Machine learning and Data science.
4. Solve optimization problems of real life using Simplex method and Assignment Problems.
5. Understand the concept of fuzzy sets and its applicability to engineering.

Unit 1: Introduction to Statistics

No of lectures – 06

Measures of central tendency, Measures of dispersion, Coefficient of variation, Central Moments, Skewness and Kurtosis.

Unit 2 : Descriptive Statistics

No of lectures – 06

Coefficient of correlation by Karl Pearson's method and lines of regression of bivariate data. Curve fitting by Least square principle (straight line, second degree parabola and exponential curves).



Unit 3: Numerical solutions to system of linear simultaneous equations No of lectures – 06

Direct methods – Gauss elimination method, Gauss – Jordan method, LU decomposition method.
Iterative methods – Gauss – Jacobi method and Gauss – Seidel method.

Unit 4: LPP and Assignment Problems

No of lectures – 06

Introduction to LPP, Simplex method for LPP, Assignment problem: Introduction mathematical formulation of Assignment Problems, Hungarian method to solve Assignment Problems.

Unit 5: Introduction to fuzzy sets

No of lectures – 06

Crispsets, Basic types of fuzzy sets, Basic concepts of fuzzy sets, Fuzzy sets vs. Crispsets, α -cut, Strong α -cut, Additional properties of α -cuts, representation of fuzzy sets and extension principle of fuzzy sets.

Internal Continuous Assessment (ICA):

ICA shall consist of a minimum of five assignments/tutorials based on the above syllabus.

Text Books:

1. Textbook of Engineering Mathematics by N.P.Bali, Manish Goyal, Laxmi Publications, New Delhi.
 2. Advanced Engineering Mathematics, Kreyzig-John Wiley & SMS, New York.
 3. Higher Engineering Mathematics, Ramana B.V., Tata McGraw Hill.
 4. Numerical Methods in Engineering and Science by B.S.Grewal, Khanna Publication, Delhi.
 5. George J Klir and BoYuan, Fuzzy sets and Fuzzy logic–PHI India.
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Reference Books:

1. Advanced Engineering Mathematics, Peter O'Neil, Cengage Learning.
2. Pundir & Pundir, Fuzzy sets and their applications–Pragati Publications.

